EMBEDDING A DIFFERENTIAL GRADED MODULE INTO A PRODUCT OF SHIFTS OF THE DUAL ALGEBRA PRESERVING COKERNELS OF THE DIFFERENTIAL

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the free-module surjection onto M-dual (parallel to the proof of lemma-good-quotient, cited in place), the double-dual embedding into the product of the I_k, the dualized two-term sequence showing each I_k is an extension of shifts of A-dual, and the functional-separation argument for property (2) through lemma-hom-into-shift-dual-free are all retained verbatim.

Editorial note. The label lemma-good-sub names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The duality functors between left and right differential graded modules. Same chapter, Section section-modules-noncommutative-differential-graded. Supplies the duality N maps to N-dual with all its properties, through which the embedding into a product of duals is built.
- Free surjections onto duals and Hom into shifted duals. Same chapter, Lemma lemma-good-quotient (its proof) and Lemma lemma-hom-into-shift-dual-free. The parallel free-module construction produces the surjection onto M-dual, and the Hom-into-shift-dual lemma turns a separating functional on the cokernel into a map to a shift of A-dual, giving property (2).

Source reference. Original reference 09KN: sources/stacks-dga/pinned-original-source.tex, source lines 2490–2490. Prerequisite pointer invoked at proof line(s) 2680; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09K4: sources/stacks-dga/pinned-original-source.tex, source lines 2366–2366. Prerequisite pointer invoked at proof line(s) 2700; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0FQD: sources/stacks-dga/pinned-original-source.tex, source lines 2306–2306. Prerequisite pointer invoked at proof line(s) 2674; it names

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a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $M \to I$ of differential graded A-modules with the following properties*

(1) *$M \to I$ is injective,*
(2) *$\mathrm{Coker}(d_M) \to \mathrm{Coker}(d_I)$ is injective, and*
(3) *I sits in an admissible short exact sequence $0 \to I' \to I \to I'' \to 0$ where I', I'' are products of shifts of $A^\vee$.*

Proof. We will use the functors $N \mapsto N^\vee$ (from left to right differential graded modules and from right to left differential graded modules) constructed in Section 0FQD and all of their properties. For every $k \in \mathbf{Z}$ let Q_k be the free left A-module with generators x, y in degrees k and $k+1$. Define the structure of a left differential graded A-module on Q_k by setting $\mathrm{d}(x) = y$ and $\mathrm{d}(y) = 0$. Arguing exactly as in the proof of Lemma 09KN we find a surjection

$$\bigoplus\nolimits_{i \in I} Q_{k_i} \longrightarrow M^\vee$$

of left differential graded A-modules. Then we can consider the injection

$$M \to (M^\vee)^\vee \to (\bigoplus\nolimits_{i \in I} Q_{k_i})^\vee = \prod\nolimits_{i \in I} I_{k_i}$$

where $I_k = Q_{-k}^\vee$ is the "dual" right differential graded A-module. Further, the short exact sequence $0 \to A[-k-1] \to Q_k \to A[-k] \to 0$ produces a short exact sequence $0 \to A^\vee[k] \to I_k \to A^\vee[k+1] \to 0$.

The result of the previous paragraph produces $M \to I$ having properties (1) and (3). To obtain property (2), suppose $\overline{m} \in \mathrm{Coker}(\mathrm{d}_M)$ is a nonzero element of degree k. Pick a map $\lambda : M^k \to \mathbf{Q}/\mathbf{Z}$ which vanishes on $\mathrm{Im}(M^{k-1} \to M^k)$ but not on m. By Lemma 09K4 this corresponds to a homomorphism $M \to A^\vee[k]$ of differential graded A-modules which does not vanish on m. Hence we can achieve (2) by adding a product of copies of shifts of $A^\vee$. □